

Physics

□ Moving charges: motion of charges in a wire

Ampere's circuital law:

□ Magnetic field due to current carrying wire of infinite length using ampere's law.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

For a circular loop of $2\pi r$ circumference, (circular wire: r)

$$B \int_0^{2\pi r} dl = \mu_0 i \quad \left[\vec{B} \cdot d\vec{l} = B dl \cos \theta = B dl \cos 0^\circ = B dl \right]$$

and B is constant for symmetric situation

$$\Rightarrow B \cdot 2\pi r = \mu_0 i$$

$$\Rightarrow \boxed{B = \frac{\mu_0 i}{2\pi r}}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

□ Ohm's law

Deduce Ohm's law using concept of Drift velocity due to external field $\vec{E}$, the force experienced by the electron.

$$\vec{F} = -q\vec{E} \text{ (by electrons)} \quad (1)$$

$$\text{Acceleration of electron, } a = \frac{\vec{F}}{m} = -\frac{q\vec{E}}{m} \quad (2)$$

$$a = \frac{v}{t} \Rightarrow \frac{v}{t} = -\frac{q\vec{E}}{m} \Rightarrow v = -\frac{q\vec{E}t}{m}$$

$$v = \frac{q\vec{E}t}{m}$$

Though in acceleration, electron moves with a constant velocity called drift velocity due to its random motion.

$$V_d = V_{avg} = \frac{v_1 + v_2 + \dots + v_n}{n}$$

$$\Rightarrow V_1 = u_1 + a t_1$$

$$V_2 = u_2 + a t_2$$

$$V_n = u_n + a t_n$$

(τ : relaxation time)

$$\therefore V_d = \frac{(u_1 + u_2 + \dots + u_n) + a(\tau_1 + \tau_2 + \dots + \tau_n)}{n}$$

$$\Rightarrow V_d = a \tau_{avg}$$

$$\Rightarrow V_d = \frac{qE}{m} \tau$$

conductor of length l

cross sectional area A

current produced

electron per unit volume

$$\therefore I = V_d q n A$$

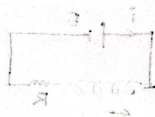
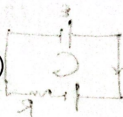
electric field produced at each point E [$V = Ed$]

$$I = \frac{qE}{m} \tau q n A = \frac{q \times V}{l m} \tau q n A$$

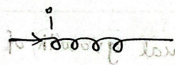
$$= \frac{n q^2 A \tau V}{l m}$$

$$\therefore V = \left(\frac{ml}{nq^2 A \lambda} \right) I$$

$$\therefore V \propto I \text{ (showed)}$$



coefficient of self inductance for a long solenoid:



N = total no of turns

l = length of solenoid

R = radius of each coil

n = number of turns per unit length $\equiv \frac{N}{l}$

B = magnetic field within the solenoid $= \mu_0 n I$

Φ linked with each coil $= BA = \mu_0 n I \pi R^2$

Total flux $\phi_T = N\Phi = N\mu_0 n I \pi R^2$

$$= (nl) \mu_0 n I \pi R^2$$

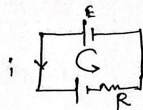
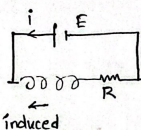
$$= \mu_0 n^2 \pi R^2 l I$$

$$\Rightarrow L I = \mu_0 n^2 \pi R^2 l I$$

$$\Rightarrow \boxed{L = \mu_0 n^2 \pi R^2 l}$$

coefficient of self inductance depends on geometry

Q4 Growth of current in L-R circuit



as inductor produces EMF we can draw a battery

when the key is closed, there is a gradual growth of current in the circuit from zero to max value

let, i be the instantaneous current at any given time t .

Then, induced emf, $E = -L \frac{di}{dt}$ because induced current is opposite to current of circuit

Applying Kirchhoff's law,

$$E - L \frac{di}{dt} - iR = 0$$

$$\Rightarrow E - iR = L \frac{di}{dt}$$

$$\Rightarrow \int_0^t \frac{dt}{L} = \int_0^i \frac{di}{E - iR}$$

$$\Rightarrow \frac{t}{L} = -\frac{1}{R} \log_e [E - iR]$$

$$\Rightarrow -\frac{Rt}{L} = \log_e \left| \frac{E - iR}{E} \right|$$

$$\Rightarrow -\frac{Rt}{L} = \log_e \left| 1 - \frac{iR}{E} \right|$$

$$\Rightarrow 1 - \frac{iR}{E} = e^{-\frac{Rt}{L}}$$

$$\Rightarrow \frac{iR}{E} = 1 - e^{-\left(\frac{R}{L}\right)t}$$

$$\Rightarrow i = \frac{E}{R} \left(1 - e^{-\left(\frac{R}{L}\right)t} \right)$$

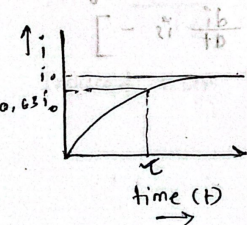
$$\Rightarrow i = i_0 (1 - e^{-\frac{Rt}{L}}) \quad (\text{Proved})$$

$$\Rightarrow i = i_0 (1 - e^{-t/\tau}) \quad \left(\tau = \frac{L}{R} \right)$$

inductive time constant

When $t = \tau$, $i = i_0 (1 - e^{-1})$

$$i = 0.63 i_0$$



at $t = 0$, $i = 0$

at $t = \infty$, steady state $i = i_0$

$$i = i_0 (1 - e^{-\infty})$$

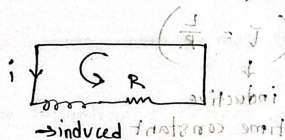
$$\left(\frac{i}{i_0} = \frac{i_0}{i_0} (1 - 0) \right)$$

(Short circuit)

$$i = i_0$$

Decay of current in LR circuit $\frac{di}{dt} = -\frac{R}{L}i$ <

After maximum current state is reached, the battery is removed from circuit. Thus, an induced emf equal to $-L \frac{di}{dt}$ is produced in Inductance L and slows down decay rate of current. (Induced current in same direction this time) <



Applying Kirchhoff's Laws

$$-L \frac{di}{dt} - iR = 0$$

$$\Rightarrow iR = L \frac{di}{dt} \Rightarrow i = i_0 e^{-\frac{R}{L}t}$$

$$i = \frac{1}{L} \int_0^t R i_0 e^{-\frac{R}{L}t} dt = i_0 e^{-\frac{R}{L}t}$$

$$\Rightarrow -\frac{Rt}{L} = \log_e \left(\frac{i}{i_0} \right)$$

$$\Rightarrow \frac{i}{i_0} = e^{-\frac{Rt}{L}}$$

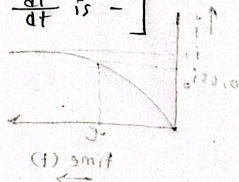
$$\Rightarrow i = i_0 e^{-\frac{Rt}{L}}$$

$$\Rightarrow i = i_0 e^{-t/\tau}$$

($i = i_0 e^{-\frac{Rt}{L}}$) $i = i_0 e^{-\frac{Rt}{L}}$ <

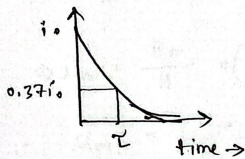
[current] decreasing

so $\frac{di}{dt}$ is -



At $t = \tau$, $i = i_0 e^{-1} = i_0 e^{(1/e)}$

$\Rightarrow i = 0.37 i_0$



Ex Mutual Inductance of two coils. One is primary and the other is secondary.

$(\phi_2)_T \propto i_1$

$(\phi_2)_T = M i_1$

$M = \frac{(\phi_2)_T}{i_1}$

Induced EMF in coil 2, $\epsilon_2 = - \frac{d(\phi_2)_T}{dt} = - \frac{d(M i_1)}{dt}$

$\epsilon_2 = - M \frac{di_1}{dt}$

$(X + H) M = B$

□ Magnetism

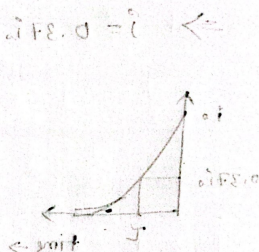
Magnetic susceptibility $\rightarrow \chi$

Magnetic flux density $\rightarrow B$

magnetization $\rightarrow M$

Magnetic field intensity $\rightarrow H$

$$B_{\text{net}} = B_{\text{ext}} + B_{\text{int}}$$



External factor: Magnetic field (external) : H

Internal factor: Due to alignment of dipoles $\rightarrow M$

$$\therefore B \propto (H + M)$$

$$\Rightarrow B = \mu_0 (H + M) \quad (\mu_0 = \text{permeability of free space})$$

$$\Rightarrow \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$

$$\frac{B}{\mu_0} = H + M$$

$$\text{Again, } B = \mu_0 (H + M)$$

$$H \propto M$$

$$M = \chi H$$

$$\chi = \frac{M}{H} \quad (\chi \rightarrow \text{susceptibility} \rightarrow \text{depends on material and temp})$$

$$\therefore B = \mu_0 (H + \chi H)$$

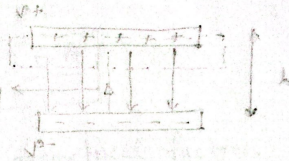
$$= \mu_0 H (1 + \chi)$$

Magnetic field applied in vacuum, $B_0 = \mu_0 H$

If we place a material,

$$B_m = \mu_m H$$

$$\Rightarrow \mu_m = \frac{B_m}{H} = (1 + \chi) \mu_0$$



$$\therefore \mu_m = \frac{\mu_m B_m}{B_m} = \frac{\mu_m B_m}{\mu_0 H} = \frac{B_m}{B_0 H} = \frac{B_m}{B_0 H}$$

$$\mu_m = (1 + \chi) \mu_0$$

$\mu_r \rightarrow$ relative permeability

Applying Gauss's law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

[$\vec{E}$ and $d\vec{A}$ are parallel]

$$\Rightarrow E \int dA = \frac{Q}{\epsilon_0}$$

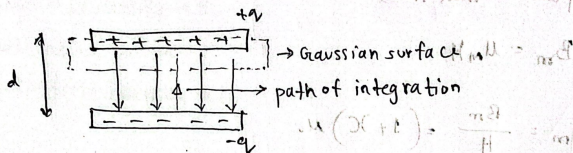
$$\Rightarrow E = \frac{Q}{\epsilon_0 A} \quad (1)$$

The potential difference is the integral of electric field over

$$V = \int_0^d \vec{E} \cdot d\vec{s} = E d \quad (2)$$

Capacitor

Parallel plate capacitor



We assume that plates of our parallel-plate capacitor are so large and so close together that we can take $\vec{E}$ to be constant throughout the region between the plates. We take a gaussian surface around the positive charge, for applying Gauss's law.

Applying Gauss's law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

$$\Rightarrow EA \cos 0^\circ = \frac{q}{\epsilon_0} \quad [\vec{E} \text{ and } d\vec{A} \text{ are parallel}]$$

$$\Rightarrow E = \frac{q}{A\epsilon_0} \quad \text{--- (1)}$$

The potential difference is the integral of electric field over path.

$$V = \int E \cdot ds = E \int_0^d ds = Ed \quad \text{--- (1)}$$

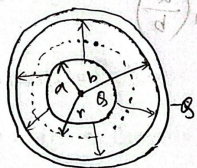
Applying (i) and (ii) in $C = \frac{q}{V}$

$$\Rightarrow C = \frac{q}{Ed} = \frac{q}{\frac{q}{\epsilon_0 A} d} = \epsilon_0 \frac{A}{d}$$

$$\therefore C = \epsilon_0 \frac{A}{d}$$

capacitance only depends on geometrical factors.

III Cylindrical capacitor



Let, a hollow cylinder has inside radius a , outer radius b and is of length l . We assume a gaussian surface on positive surface of the capacitor of radius r .

Applying Gauss's law

$$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

$$\Rightarrow E = \frac{q}{A \epsilon_0} = \frac{q}{2\pi r l \epsilon_0} \quad \text{--- (1)}$$

[Here, $A = 2\pi r l$ because only the flux of surface is considered, not that of the end caps]

Potential difference

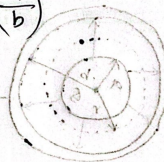
$$V = \int_{-}^{+} E \cdot ds = - \int_b^a E \cdot dr$$

$$= - \frac{q}{\epsilon_0 2\pi l} \int_b^a \frac{dr}{r}$$

$$= - \frac{q}{\epsilon_0 2\pi l} [\ln r]_b^a$$

$$= - \frac{q}{\epsilon_0 2\pi l} \ln\left(\frac{a}{b}\right)$$

$$V = \frac{q \cdot \ln\left(\frac{b}{a}\right)}{\epsilon_0 2\pi l}$$



Now, $C = \frac{q}{V} = \frac{2\pi\epsilon_0 l E \times \epsilon_0}{V}$

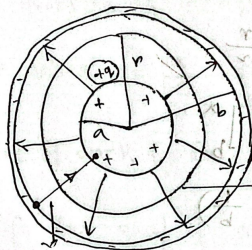
$$\Rightarrow \frac{q}{V} = \frac{2\pi\epsilon_0 \times l}{\ln b/a}$$

$$C = 2\pi\epsilon_0 \cdot \frac{l}{\ln b/a}$$

$$\frac{V}{E} = \frac{q}{\epsilon_0 E}$$

$$(1) \quad \frac{V}{E} = \frac{q}{\epsilon_0 E}$$

Spherical Capacitor



path of integration

Let, a capacitor consists of two concentric spherical shells of radii a and b . As a Gaussian surface, we draw a sphere of radius r concentric with the shells.

Applying Gauss law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\Rightarrow EA = \frac{q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi r^2 \epsilon_0} q$$

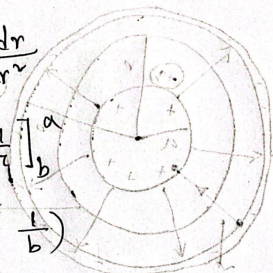
Potential difference, $V = \int E \cdot ds$

$$= - \int_b^a E dr$$

$$= - \frac{q}{4\pi\epsilon_0} \int_b^a \frac{dr}{r^2}$$

$$= - \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_b^a$$

$$= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$



$$\Rightarrow V = \frac{q}{4\pi\epsilon_0} \left(\frac{b-a}{ab} \right)$$

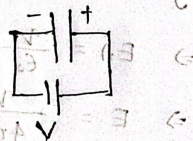
$$\therefore \frac{q}{V} = \frac{4\pi\epsilon_0 (ab)}{b-a}$$

$$\therefore \boxed{C = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right)}$$

Energy Stored in a Capacitor

At, at $t=0$, $q=0$, $V=0$

at $t=t'$, $q=q'$, $V = \frac{q'}{C}$



suppose, at any instant t' , a charge q' has been transferred

from one plate to another, So, the potential difference, $V' =$

We want to transfer dq' charge, so work done will

$$\text{be } dW = V' dq' = \frac{q'}{C} dq'$$

Total amount of work to accumulate all the charges,

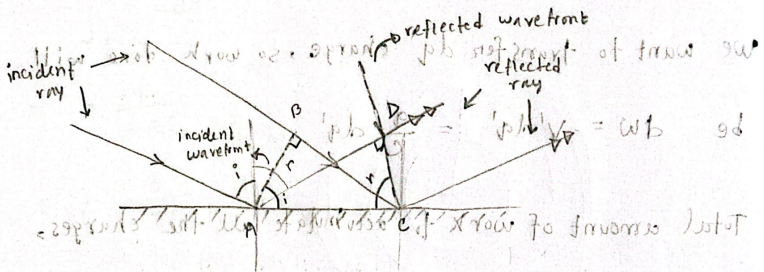
$$W = \int_0^q dW = \frac{1}{C} \int_0^q q' dq' = \frac{1}{C} \left[\frac{q'^2}{2} \right]_0^q = \frac{q^2}{2C}$$

Work done is potential energy of the capacitor

$$U = \frac{q^2}{2C} = \frac{1}{2} CV^2$$

PHYSICAL OPTICS

Proof of Reflection using Huygen's principle



Two incident rays on the surface. We draw a wavefront at time t . The rays of light are perpendicular to the wavefront according to Huygen's principle. Taking radius equal to BC we draw an arc from point A . The tangent to this from C is the reflected wavefront. We draw the reflected rays perpendicular to the wavefront.

According to the figure,

[i = angle of incidence
 r = angle of reflection]

$$\angle BAC = i$$

$$\text{and } \angle ACD = r$$

Again, in $\triangle ABC$ and $\triangle ACD$,

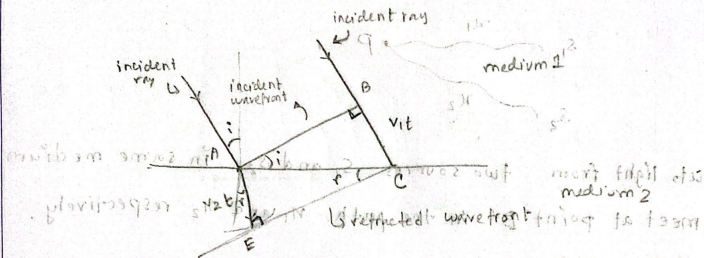
AC is common

$$\angle ABC = \angle ADC \text{ [perpendiculars]}$$

and $AD = BC$ [the rays will take equal time to cover the distance s]

$\therefore i = r$ (proved)

Proof of refraction using Huygen's Principle:



u_1 , v_1 and v_2 represent speed of light in medium 1 and medium 2 respectively. u_1 , wavefront AB incident on the interface at angle i . If t is the time taken by (wavefront) to travel distance BC, $BC = v_1 t$.

We draw a sphere of radius $v_2 t$ from point A in second medium. CE is tangent drawn from C on the sphere. CE is the refracted wavefront.

For $\triangle ABC$ and $\triangle AEC$

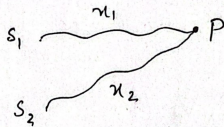
$$\frac{\sin i}{\sin r} = \frac{\frac{BC}{AC}}{\frac{AE}{AC}} = \frac{(v_1 t)}{v_2 t}$$

$$\therefore \frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

For medium 1, $n_1 = \frac{c}{v_1}$
 For medium 2, $n_2 = \frac{c}{v_2}$

$$\therefore n_1 \sin i = n_2 \sin r \quad [\text{Snell's law}]$$

Ex. Phase difference = $\frac{2\pi}{\lambda} \times \text{Path difference}$



Let light from two sources S_1 and S_2 in same medium meet at point P in the path x_1 and x_2 respectively.

Here, $f \rightarrow$ same

$$\omega = 2\pi f \rightarrow \text{same}$$

$$v \rightarrow \text{same}$$

$$k = \frac{2\pi}{\lambda} \text{ (wave factor)} \rightarrow \text{same}$$

only path is different.

$$\text{We know, } y = a \sin(\omega t + kx)$$

where a is amplitude

and $(\omega t + kx)$ is the phase

$$\text{Now, } y_1 = A_1 \sin(\omega t + kx_1)$$

$$y_2 = A_2 \sin(\omega t + kx_2)$$

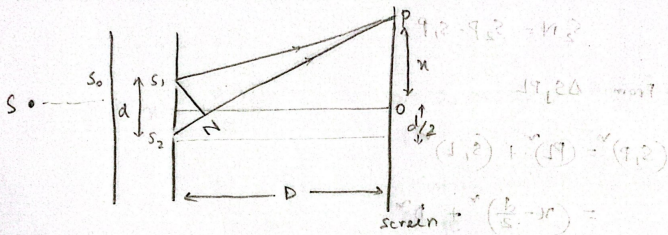
$$\therefore \text{Phase difference, } \phi = \omega t + kx_2 - \omega t - kx_1$$

$$= k(x_2 - x_1)$$

$$= k \Delta x$$

$$\therefore \text{Phase difference} = \frac{2\pi}{\lambda} \times \text{Path difference}$$

Young's Double Slit Experiment : Finding out Path Difference



Suppose, S is monochromatic source of light. S_0 is the slit through which light passes and illuminates the slits S_1 and S_2 . Waves emitted by slits S_1 and S_2 are part of the same wave-front, so these waves have the same frequency and the same phase.

Hence, slits S_1 and S_2 behave as two coherent sources. Interference takes place on screen. If we consider a point O on perpendicular bisector of S_1S_2 , waves traveling along S_1O and S_2O have travelled equal distances. Hence, they arrive at O in phase and interfere constructively to make O the center of a bright fringe or maxima.

To locate position of maxima and minima on both sides of O , we consider any point P at x distance from O . S_1P and S_2P are joined. We draw a normal S_1N on S_2P . Then,

Path difference between S_2P and S_1P :

$$S_2N = S_2P - S_1P$$

From ΔS_1PL

$$\begin{aligned}(S_1P)^r &= (PL)^r + (S_1L)^r \\ &= \left(x - \frac{d}{2}\right)^r + D^r\end{aligned}$$

From ΔS_2PM

$$\begin{aligned}(S_2P)^r &= (MP)^r + (S_2M)^r \\ &= \left(x + \frac{d}{2}\right)^r + D^r\end{aligned}$$

$$\text{Now, } (S_2P)^r - (S_1P)^r = \left(x + \frac{d}{2}\right)^r - \left(x - \frac{d}{2}\right)^r$$

$$= x^r + \frac{d^r}{4} + 2xd - x^r + \frac{d^r}{4}$$

$$\therefore (S_2P + S_1P)(S_2P - S_1P) = 2xd$$

$$\Rightarrow S_2P - S_1P = \frac{2xd}{2D}$$

[since the distance of screen from slits

S_1 and S_2 is very large, so $S_2P \approx S_1P \approx D$]

$$\therefore \text{Path difference} = \frac{nd}{D}$$

For bright fringe

($\frac{x d}{D}$ is integral multiple of λ)

$$\frac{x d}{D} = p \lambda$$

$$\Rightarrow x = \frac{p \lambda D}{d}$$

$p = 0, x = 0$, central fringe bright

$$p = 1, x_1 = \frac{\lambda D}{d}$$

(that is, position of first bright fringe)

Now, $x_p = \frac{p \lambda D}{d}$

$$x_{p+1} = \frac{(p+1) \lambda D}{d}$$

$\therefore$ Distance between two successive bright fringes,

fringe width, $\beta = x_{p+1} - x_p$

$$= \frac{(p+1) \lambda D}{d} - \frac{p \lambda D}{d}$$

$$\beta = \frac{\lambda D}{d}$$

For dark fringe

($\frac{x d}{D}$ is odd multiple of $\lambda/2$)

$$\therefore \frac{x d}{D} = (2p+1) \frac{\lambda}{2}$$

$$\Rightarrow x = \frac{(2p+1) \lambda D}{2d}$$

$$\text{Now, } x_p = \frac{(2p+1)\lambda D}{2d}$$

$$x_{p+1} = \frac{(2p+3)\lambda D}{2d}$$

$$\therefore \beta = \frac{x_{p+1} - x_p}{x_p}$$

$$= \frac{(2p+3)\lambda D}{2d} - \frac{(2p+1)\lambda D}{2d}$$

$$\beta = \frac{\lambda D}{d}$$

$$x_p = \frac{b x}{D}$$

$$\frac{p \lambda D}{b} = x >$$

$$\frac{p \lambda D}{b} = x_p$$

$$\frac{(p+1)\lambda D}{b} = x_{p+1}$$

Distance between two successive bright fringes

$$\text{fringe width } b = x_{p+1} - x_p$$

$$= \frac{(p+1)\lambda D}{b} - \frac{p \lambda D}{b}$$

$$b = \frac{\lambda D}{d}$$

$$\left(\frac{D}{d} \text{ is a multiple of } \lambda \right)$$

dark fringe

$$\frac{x}{b}$$

$$(2p+1)$$

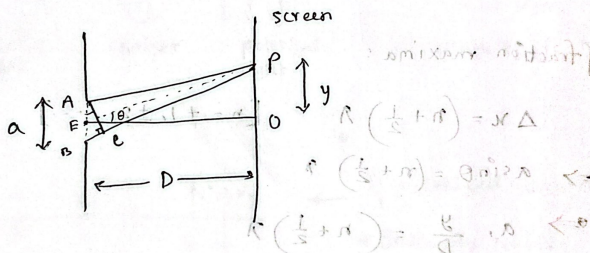
Diffraction : $\sin \theta = \frac{\lambda}{a}$

Single slit diffraction:

Slit AB of width a . Distance between slit and screen is D .

At point P on the screen, secondary waves interfere destructively and produce a dark fringe.

Let, y is the distance between point P and Point O (center of screen). We draw $Ac \perp BP$. Let, θ is the angle of diffraction.



Here, $D \gg a$.

$$\therefore \angle BAC = \theta$$

$$\text{and } \sin \theta \approx \tan \theta \approx \theta = \frac{y}{D} \quad (\text{in } \triangle POE)$$

The path difference between rays AP and BP is

$$\Delta x = BP - AP = BC$$

$$\text{In } \triangle BCA, \sin \theta = \frac{BC}{AB} \Rightarrow \sin \theta = \frac{BC}{a}$$

$$\Rightarrow BC = a \sin \theta \Rightarrow \boxed{\Delta x = a \sin \theta}$$

Diffraction minima :

$$\Delta x = n\lambda \quad (n = \pm 1, \pm 2, \pm 3)$$

$$a \sin \theta = n\lambda$$

$$\text{or, } a \cdot \frac{y}{D} = n\lambda$$

$$\text{or, } y_n = \frac{n\lambda D}{a}$$

Distance of n th dark fringe from the center

$$\sin \theta \approx \theta$$

$$\Rightarrow \lambda = a \sin \theta \quad (n=1)$$

$$\Rightarrow \lambda = a\theta$$

$$\Rightarrow \theta = \frac{\lambda}{a}$$

$$\text{Again, } \theta = \frac{y}{D}$$

$$\therefore \frac{\lambda}{a} = \frac{y}{D}$$

$$\Rightarrow y = \frac{\lambda D}{a} \quad (n=1)$$

Diffraction maxima :

$$\Delta x = \left(n + \frac{1}{2}\right) \lambda$$

$$\Rightarrow a \sin \theta = \left(n + \frac{1}{2}\right) \lambda$$

$$\Rightarrow a \cdot \frac{y}{D} = \left(n + \frac{1}{2}\right) \lambda$$

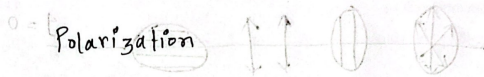
$$\Rightarrow \frac{ay}{D} = \left(n + \frac{1}{2}\right) \lambda$$

$$\Rightarrow y_n = \frac{\left(n + \frac{1}{2}\right) \lambda D}{a}$$

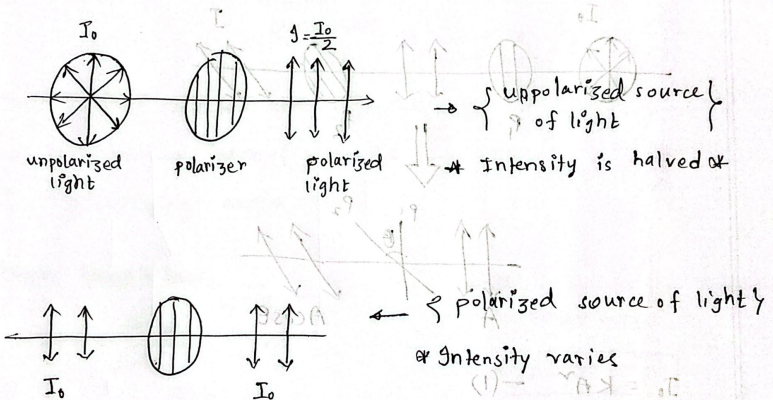
Distance of n th bright fringe from center

$$\frac{ay}{D} = \left(n + \frac{1}{2}\right) \lambda$$

Width of central maxima = $\frac{2\lambda D}{a}$ [on both sides]



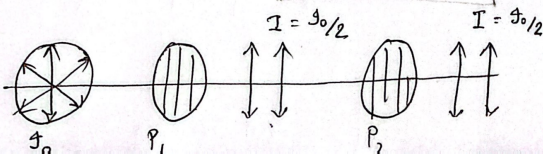
Q. How can you distinguish betn unpolarized light and polarized light using polaroid?



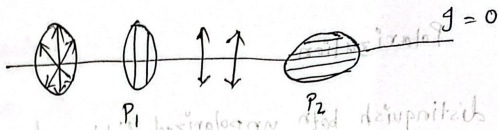
Q. Malu's law

Intensity of polarized light when it passes through the polaroid :

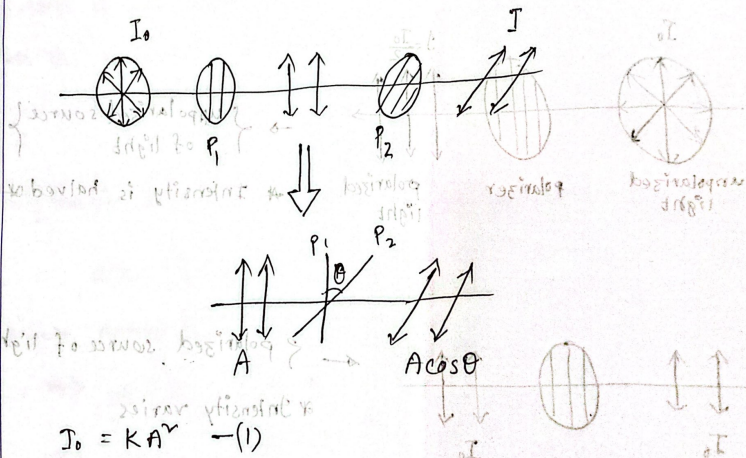
Case 1 : $\theta = 0^\circ$ (θ = angle between the two polaroids P_1 and P_2)



Case 2, ($\theta = 90^\circ$)



Case 3 (angle θ)



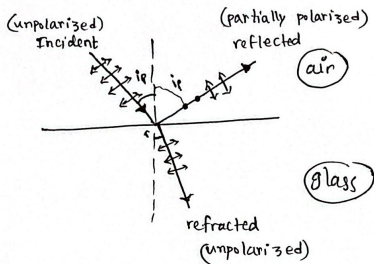
$$I_0 = KA^2 \quad (1)$$

Again, $I \propto (A \cos \theta)^2$

$$\Rightarrow I = KA^2 \cos^2 \theta$$

$$\Rightarrow \boxed{I = I_0 \cos^2 \theta} \quad [\text{from (1)}]$$

☐ Brewster's law



- horizontal component
↔ vertical component

Angle between reflected and refracted ray is 90°

μ is refractive index of material (say glass)

i_p is polarizing angle

From Snell's law,

$$\mu = \frac{\sin i_p}{\sin r}$$

$$\Rightarrow \mu = \frac{\sin i_p}{\sin(90^\circ - i_p)}$$

$$\Rightarrow \mu = \frac{\sin i_p}{\cos i_p}$$

$$\Rightarrow \boxed{\mu = \tan i_p}$$

Hence,

$$i_p + 90^\circ + r = 180^\circ$$

$$\Rightarrow \boxed{r = 90^\circ - i_p}$$

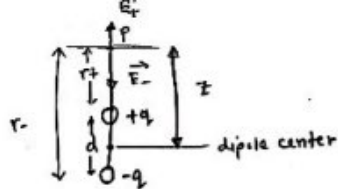


Figure shows two charges $+q$ and $-q$ separated by d .

Applying principle of superposition for electric field, $E = E_+ - E_-$

$$E = E_+ - E_-$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r_+^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_-^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_+^2} - \frac{1}{r_-^2} \right)$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{(z - d/2)^2} - \frac{1}{(z + d/2)^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left\{ \frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left[\frac{1 + \frac{d^2}{4z^2} + \frac{d}{z} - 1 + \frac{d}{z} - \frac{d^2}{4z^2}}{\left(1 - \frac{d}{2z}\right)\left(1 + \frac{d}{2z}\right)} \right]$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \frac{2d/z}{\left(1 - \frac{d^2}{4z^2}\right)^2}$$

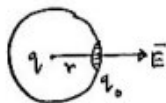
Considering $z \gg d$ we have $d/2z \ll 1$.

Then we can neglect $d/2z$ in the former equation and,

$$E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3}$$

Proof of Gauss' law for Coulomb's law:

We take a charge q enclosed by a spherical gaussian surface.
we take small segment dA . Area vector is in same direction
as the field $\vec{E}$.



We place test charge q_0 in dA .

Coulomb's law: $F = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2}$

$$E = F/q_0 = \frac{q}{4\pi\epsilon_0 r^2} \quad (1)$$

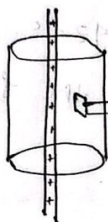
$$\text{Gauss } \Phi = \oint \vec{E} \cdot d\vec{A}$$

$$= \oint E dA \cos \theta = E \oint dA$$

$$= E \cdot 4\pi r^2 = \frac{q}{4\pi\epsilon_0 r^2} \cdot 4\pi r^2$$

$$\therefore \boxed{\Phi = \frac{q}{\epsilon_0}}$$

Application of Gauss's theorem (Cylindrical symmetry)



$$q = EA \cos \theta$$

$$= EA$$

$$= E \times 2\pi rh$$

$$\theta = 0^\circ$$

$$\epsilon_0 q = q_{\text{enc}}$$

$$\Rightarrow \epsilon_0 \cdot E \cdot 2\pi rh = q_{\text{enc}}$$

$$\Rightarrow \epsilon_0 \cdot E \cdot 2\pi rh = \lambda \times h$$

$$\Rightarrow E = \frac{\lambda}{2\pi \epsilon_0 r}$$

End cap have no flux
because $\vec{E} \perp$ Area vec
 $\cos 90^\circ = 0$.

** draw it
my
for deriva

$\lambda \rightarrow$ linear charge density

* Electric field intensity due to thin infinite plane sheet:

$\sigma \rightarrow$ surface charge density.

$$\epsilon_0 \oint E \cdot dA = q_{\text{enc}}$$

$$\Rightarrow \epsilon_0 (EA + EA) = \sigma A$$

$$\Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

$$\sigma \rightarrow \frac{q}{A} \text{ (per unit area)}$$

$$\lambda \rightarrow \frac{q}{d} \text{ (per unit length)}$$